

Question	Marking Guidance	Mark	Comments
11.1	1. Helicase; 2. Breaks hydrogen bonds; 3. Only one DNA strand acts as template; 4. RNA nucleotides attracted to exposed bases; 5. (Attraction) according to base pairing rule; 6. RNA polymerase joins (RNA) nucleotides together; 7. Pre-mRNA spliced to remove introns;	6 max	
11.2	1. Polymer of amino acids; 2. Joined by peptide bonds; 3. Formed by condensation; 4. Primary structure is order of amino acids; 5. Secondary structure is folding of polypeptide chain due to hydrogen bonding; 6. Tertiary structure is 3-D folding due to hydrogen bonding <u>and</u> ionic/disulfide bonds; 7. Quaternary structure is two or more polypeptide chains;	5 max	5. Accept alpha helix/pleated sheet
11.3	1. Hydrolysis of peptide bonds; 2. Endopeptidases break polypeptides into smaller peptide chains; 3. Exopeptidases remove terminal amino acids; 4. Dipeptidases hydrolyse/break down dipeptides into amino acids;	4	

